UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	NOVEMBER 2026
TIME	:	

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). Generated answers show working. Real past-paper questions are reproduced with their source reference; no answer is invented where the original scheme is unavailable.

2. Generated marking points state the expected answer and the allocation of marks. Real-bank questions retain their source paper and page reference.

3. Part A = 4 x 5 + 6 x 10; Part B = 1 x 20; total = 100 marks.

4. Generated slots: 2. Real-bank slots: 9. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 20 unique CSC429 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

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PART A — MARKING GUIDE

QUESTION A1 (5 marks)

Topic: Instruction formats and addressing modes

Given the following Figure 2.
Instruction
Opcode
Address A
Memory
Operand
Figure 2
Label the addressing technique illustrated in Figure 2.
a)
Explain how the addressing technique in 9(a) is used.
b)
List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

Source: CSC429 (1).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A2 (5 marks)

Topic: Data representation and number systems

Calculate -10 - 74 using the sign-and-magnitude method. Assume an 8-bit byte. Show your complete workflow.

Answer: left -10 -> 1 0001010; right 74 -> 0 1001010; -10 - 74 = -84 -> sign-magnitude 1 1010100

- *1 mark for converting both operands to sign-magnitude.*
- *1 mark for applying the sign rule for subtraction.*
- *1 mark for the correct magnitude operation.*
- *1 mark for the correct sign bit.*
- *1 mark for the final decimal and byte representation.*

QUESTION A3 (5 marks)

Topic: Parallel and multiprocessor architecture

Given the following Non Uniform Memory Access (NUMA) diagram.
2-1
1
'2
Figure 3: NUMA Architecture
a) State ONE (1) advantage of Symmetric Multi-Processor (SMP) that is applied in NUMA architecture.
b) Describe ONE (1) disadvantage of cluster architecture.
c) Draw a possible architecture for cluster architecture based on Figure 3.

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A4 (5 marks)

Topic: Computer generations and history

Question 1 is based on the following snippet garnered from a computer science web page.
Henry Edward "Ed" Roberts
(September 13, 1941 -April 1, 2010)
was an American engineer, entrepreneur and
medical doctor who invented the first
commercially successful personal computer in

1975. He is most often known as "the father of the personal computer". The computer was available as a kit for $439 or assembled for $621 and had several additional add-ons such as a memory board and interface boards. By August 1975, over 5,000 Altair 8800 personal computers were sold; starting the personal computer revolution.

Figure 1: Snippet from a computer science web page.

a) In which computer generation that a personal computer is introduced?
b) Discuss ONE (1) other major discovery that is found in this generation.
c) Advances in laptop and hand-held devices are accelerating during this computer generation. How such technology is feasible as compared to previous computer generation?

Source: CSC429 (8).pdf, page(s) 2

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A5 (10 marks)

Topic: Input/output and interrupts

a) Explain the problem in polling 1/0 method.
b) Explain how buffering can help to improve the device overall performance?
c) Discuss how DMA technique can help the 1/0 transfer performance instead of using polling method and interrupt 1/0 method?

Source: CSC429 (10).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A6 (10 marks)

Topic: Computer architecture and CPU organisation

a) Discuss the whole operation of the machine cycle and its respective responsible computer components.
b) Elaborate on what will happen in the machine cycle if the main memory is corrupted.

Source: CSC429 (21).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A7 (10 marks)

Topic: Input/output and interrupts

Question 5 is based on the following scenario.

"Nurathia Farhana browses the UiTM4fly.com website to book for a return flight from KLIA to LAX. While comparing for a better price, two messages popped up in her screen. Figure 1 (a) and (b) illustrate the conveyed messages.

A hardware failure has been found. Please restart your computer now.

(a)

Do you want to be upgraded to business class now?

(b)

Figure 1: The pop-up messages

a) Which message is the non-maskable interrupt? Justify your answer by giving appropriate reason.
b) Differentiate between maskable and non-maskable interrupt?
c) Outline the common way to handle multiple interrupts as happened in the given scenario.

Source: CSC429 (11).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source

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paper and its mark allocation; no answer has been invented.

QUESTION A8 (10 marks)

Topic: Memory hierarchy and cache

Replacement algorithm is used to determine which entry in a cache to flush to main memory when it is desired to cache a new block of data.
a) Which mapping function does not need to have a replacement algorithm?
b) Describe FOUR (4) types of replacement algorithm.

Source: CSC429 (5).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A9 (10 marks)

Topic: Computer architecture and CPU organisation

a) Compare system interconnection and CPU interconnection in terms of:
i) function
ii) physical appearance.
b) Describe the function of the THREE (3) different bus types in a computer system.

Source: CSC429 (16).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A10 (10 marks)

Topic: Data representation and number systems

Answer both parts. Show your complete workflow.
a) Convert 314510 to hexadecimal.
b) Convert ED116 to decimal.

a) Convert 314510 to hexadecimal.

Answer: 314510 = C4916

- *1 mark for repeated division by 16.*
- *1 mark for correct remainders.*
- *1 mark for reading remainders in reverse order.*
- *1 mark for the correct hexadecimal digits.*
- *1 mark for the base subscript.*

b) Convert ED116 to decimal.

Answer: ED116 = 379310

- *1 mark for place values 16^2, 16^1 and 16^0.*
- *1 mark for correct digit values.*
- *1 mark for correct products.*
- *1 mark for correct addition.*
- *1 mark for the final decimal value.*

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PART B — MARKING GUIDE

QUESTION B1 (20 marks)

Topic: Assembly-language programming

Question 1 is based on Figure 1 and Table 1.
z=A2-B*C+jl_
D+5
Figure 1
write an appropriate program using correct assembly language notations to compute the equation snbwn in Figure 1 using instructions for zero-address, one-address, and three-address. The available instructioni are provided in Tabte 1. The register labeled A, B, C, and D are fixed, white registers x, y, and z can be used temporarily. The final answer must be stored in register z.
Table 1: Machine instruction
Three-address
One-address
Zero-address
ADD (p,q,r)
LOAD p
PUSH p
suB (p,q,r)
STOR p
POP p
MUL (p,q,r)
ADD p
ADD
DIV (p,q,r)
SUB p
SUB
MUL p
MUL
DIV p
DIV

Source: CSC429 (20).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

END OF MARKING SCHEME

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